

SOTKAMO KUOLANIEMI, Finland

WMO: 027390

Lat: 64.112N

Lon: 28.336E

Elev: 163

StdP: 99.39

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.4	(c) -26.6	(d) -32.3	(e) 0.2	(f) -29.3	(g) -29.3	(h) 0.3	(i) -26.6	(j) 7.5	(k) -5.4	(l) 7.1	(m) -5.6	(n) 0.9	(o) 160	(p) 0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.5	17.9	24.4	16.9	22.5	16.0	19.5	23.8	18.4	22.7	17.2	21.0	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.3	21.2	16.7	12.1	20.0	15.5	11.2	18.9	56.3	23.6	52.5	22.7	48.8	21.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.8	6.1	5.4	-32.4	28.1	4.3	3.3	-35.6	30.5	-38.1	32.5	-40.5	34.3	-43.7	36.7
			-32.2	21.3	4.2	2.0	-35.2	22.7	-37.7	23.9	-40.0	25.1	-43.0	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-11.1	-9.0	-4.6	1.8	8.7	12.9	16.4	14.3	9.6	2.6	-1.8	-6.7		
	DBStd	10.50	7.99	7.94	5.46	3.99	4.85	3.94	3.52	3.21	3.19	3.58	4.46	6.99		
	HDD10.0	3138	654	532	453	247	84	15	1	4	46	231	353	517		
	HDD18.3	5697	913	765	712	497	301	168	81	131	263	488	603	776		
	CDD10.0	516	0	0	0	1	42	103	199	137	33	1	0	0		
	CDD18.3	33	0	0	0	0	1	6	21	6	0	0	0	0		
	CDH23.3	288	0	0	0	0	22	61	168	37	0	0	0	0		
	CDH26.7	43	0	0	0	0	1	7	31	4	0	0	0	0		
(14) Wind	WSAvg	2.8	2.7	2.8	3.0	2.8	2.9	2.9	2.4	2.5	2.7	2.9	2.9	2.9		
(15) Precipitation	PrecAvg	676	45	39	35	31	58	71	89	75	64	62	54	53		
	PrecMax	865	75	67	52	59	108	152	175	164	100	124	109	95		
	PrecMin	552	11	7	10	9	10	28	29	24	13	17	6	14		
	PrecStd	77	16	19	11	13	30	30	33	37	25	27	26	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	4.6	8.1	17.2	25.6	27.8	29.2	26.8	20.2	12.3	8.6	3.7		
		MCWB	1.5	2.7	3.4	9.1	13.9	17.8	19.7	19.1	15.3	10.6	7.7	3.3		
	2%	DB	0.8	2.9	5.1	12.3	23.4	25.2	27.3	23.9	17.4	10.4	6.5	2.3		
		MCWB	0.3	1.6	1.9	6.7	13.7	17.3	19.0	17.3	13.4	9.0	5.8	1.6		
	5%	DB	0.0	1.5	3.3	9.9	20.7	22.4	25.1	22.0	15.7	9.0	4.8	1.3		
		MCWB	-0.2	0.7	0.9	5.2	12.7	15.7	18.0	16.4	12.7	7.9	4.4	0.8		
	10%	DB	-0.9	0.4	1.9	7.8	17.2	20.3	23.1	20.1	14.4	7.6	3.3	0.5		
		MCWB	-1.2	-0.1	0.4	3.8	10.6	14.1	16.9	15.6	12.1	6.7	3.0	0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	3.1	3.9	9.6	15.9	19.3	21.6	20.5	15.7	11.2	8.1	3.5		
		MCDB	1.9	4.3	6.6	14.7	22.9	23.9	26.1	25.0	19.0	11.8	8.9	3.6		
	2%	WB	0.4	1.8	2.4	7.3	14.5	18.1	20.2	18.4	14.4	9.4	5.7	1.7		
		MCDB	0.7	2.5	4.2	11.8	20.8	23.9	24.6	22.1	16.5	10.2	6.2	2.1		
	5%	WB	-0.2	0.8	1.5	5.4	13.1	16.4	19.1	17.1	13.2	8.0	4.4	0.8		
		MCDB	0.1	1.3	2.8	8.8	19.3	21.1	23.2	20.4	15.0	8.7	4.8	1.1		
	10%	WB	-1.3	0.0	0.7	4.3	11.5	14.9	18.0	16.1	12.5	6.8	3.1	0.2		
		MCDB	-1.0	0.4	1.6	6.9	17.1	19.0	22.1	19.3	14.1	7.6	3.3	0.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.6	8.3	8.4	9.4	8.9	8.8	8.1	6.4	4.4	3.6	5.0		
		MCDBR	3.8	4.6	7.6	12.1	14.9	12.3	12.1	11.0	8.3	5.8	4.4	3.6		
	5% WB	MCWBR	3.8	4.2	5.6	7.0	7.1	5.6	5.2	5.2	5.1	4.9	4.3	3.5		
		MCWBR	3.8	4.8	6.0	10.4	13.5	11.1	10.9	9.4	7.3	5.6	4.3	3.5		
(40) Clear-Sky Solar Irradiance	taub		0.213	0.276	0.305	0.342	0.344	0.344	0.368	0.348	0.325	0.303	0.363	0.204		
	taud		1.962	2.107	2.157	2.266	2.453	2.478	2.427	2.468	2.521	2.386	2.662	1.963		
	Ebn at Noon		348	622	771	830	868	875	838	824	772	613	328	137		
	Edh at Noon		27	63	92	104	95	95	98	86	66	49	25	11		
(44) All-Sky Solar Radiation	RadAvg		0.15	0.68	2.12	3.69	4.67	5.21	4.90	3.69	2.02	0.74	0.17	0.06		
	RadStd		0.02	0.15	0.22	0.40	0.51	0.35	0.51	0.38	0.22	0.12	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page